EXPERIMENT 6: ERROR DETECTION AND CORRECTION USING HAMMING CODE

AIM

To implement error detection and correction using Hamming Code with Python sender and receiver programs.

DESCRIPTION

Hamming Code is an error detection and correction technique that uses parity bits to detect and correct single-bit errors during data transmission.

SENDER PROGRAM

sender.py - C:/Users/sweth/Downloads/sender.py (3.13.5)

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```
def text_to_binary(txt):
    return ''.join(format(ord(c), '08b') for c in txt)

def hamming_encode(data):
    m = len(data)
    r = 0
    while (2 ** r) < (m + r + 1):
        r += 1

    code = ['0'] * (m + r)
    j = 0

    for i in range(1, m + r + 1):
        if i & (i - 1):
            code[i - 1] = data[j]
            j += 1

    for p in range(r):
        pos = 2 ** p
        parity = 0
        for i in range(1, len(code) + 1):
            if i & pos:
                parity ^= int(code[i - 1])
        code[pos - 1] = str(parity)

    return ''.join(code)

text = input("Enter data: ")
binary = text_to_binary(text)
encoded = hamming_encode(binary)

print("Binary data:", binary)
print("Hamming code:", encoded)

with open("channel.txt", "w") as f:
    f.write(encoded)

print("Data saved to channel.txt")
```

RECEIVER PROGRAM

receiver.py - C:/Users/sweth/Downloads/receiver.py (3.13.5)

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```
    error_pos = 0

    for p in range(r):
        pos = 2 ** p
        parity = 0

        for i in range(1, len(code) + 1):
            if i & pos:
                parity ^= int(code[i - 1])

        if parity:
            error_pos += pos

    if error_pos:
        code[error_pos - 1] = str(
            1 - int(code[error_pos - 1])
        )
        print("Error corrected")
    else:
        print("No error detected")

    data = ""
    for i in range(1, len(code) + 1):
        if i & (i - 1):
            data += code[i - 1]

    return data

with open("channel.txt", "r") as f:
    received = f.read().strip()

data = hamming_decode(received)

output = ""
for i in range(0, len(data), 8):
    output += chr(int(data[i:i + 8], 2))

print("Received Data:", output)
```

OUTPUT

IDLE Shell 3.13.5

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```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\sweth\Downloads\sender.py ==========
Enter data: HELLO
Binary data: 0100100001000101010011000100110001001111
Hamming code: 0101100010000100001010100110001100110001001111
Data saved to channel.txt
>>>
========== RESTART: C:/Users/sweth/Downloads/receiver.py ==========
No error detected
Received Data: HELLO
>>>
```

RESULT

Hamming Code was successfully implemented using Python sender and receiver programs. The data was encoded, transmitted through a file, and decoded successfully at the receiver.